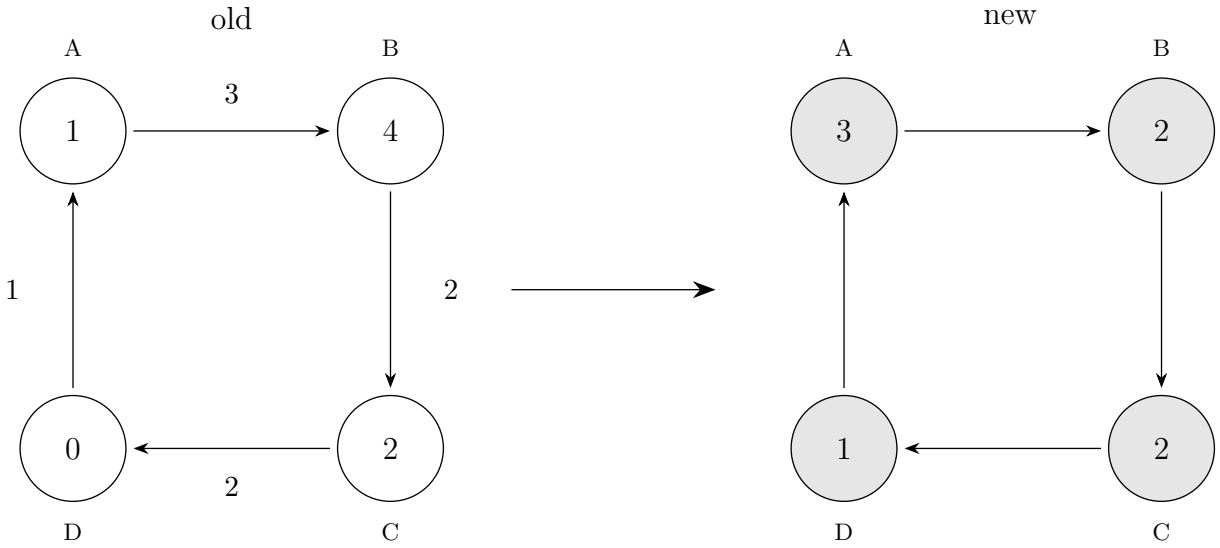
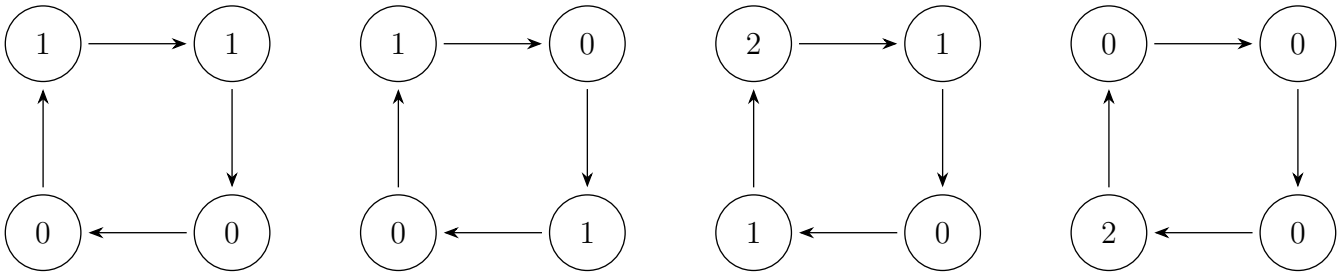
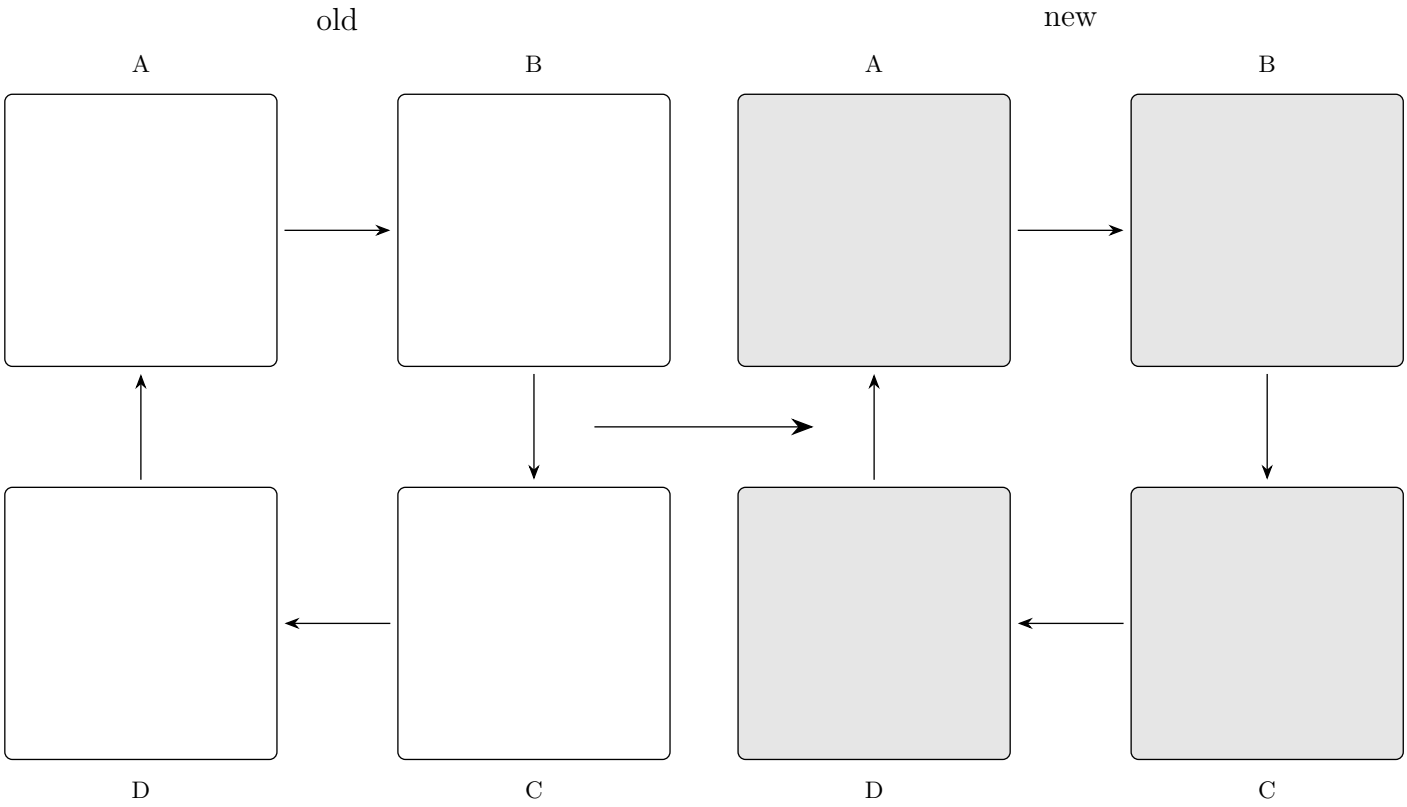
Stand the two old towers on an arrow side by side. Build the extra part on the new mat, at the arrow's start. Keep the old towers until all four new towers are built. Then clear the old mat, move the new towers onto it, and drop a counter in the cup.



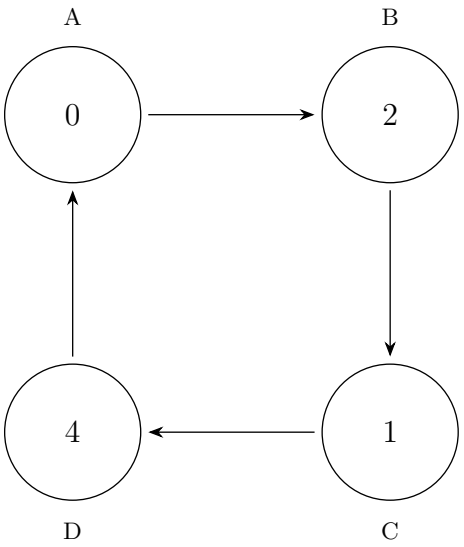
A has 1 and B has 4. The extra part is 3, so new A is 3.

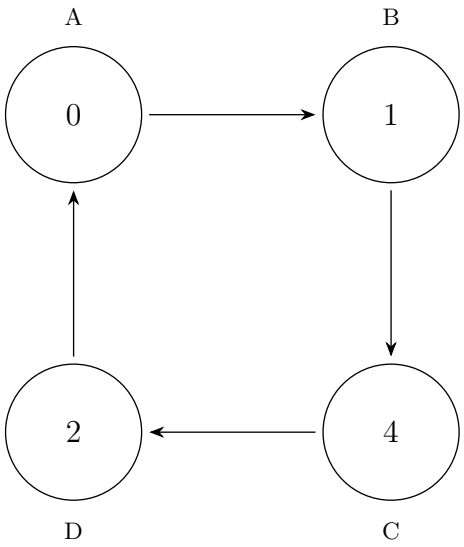
Problem 1: Build these starts and keep making new rings. Stop when all four towers are 0.

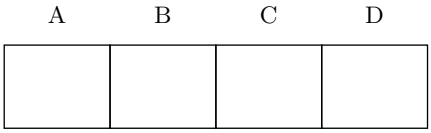




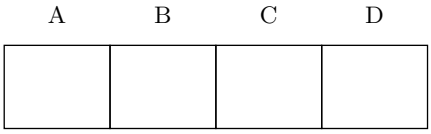
Problem 2: Put 0, 1, 2, and 4 around the four stations. Find an order that lasts longer than the other orders you try.



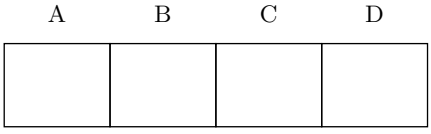




Moves to zero: _____

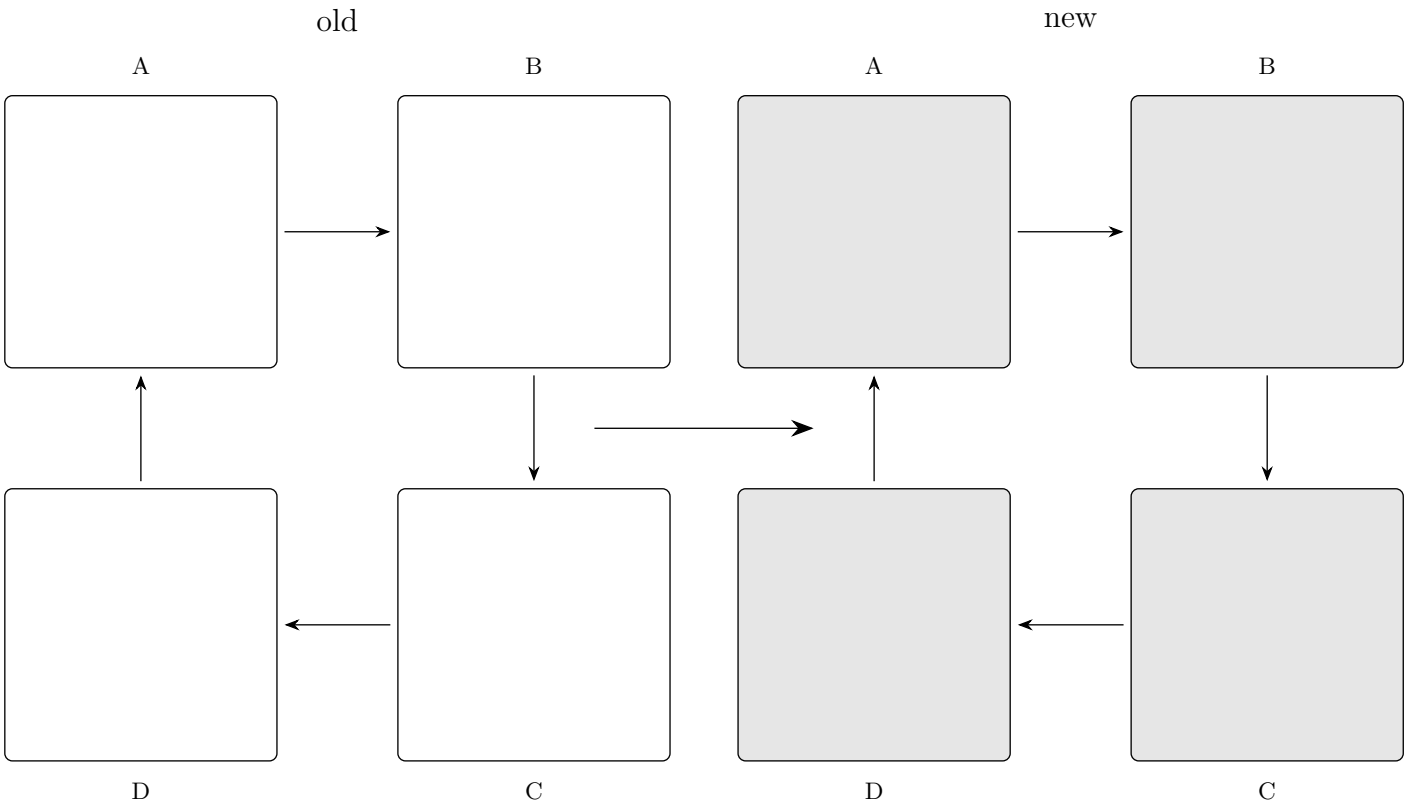


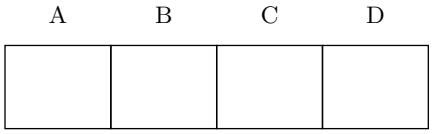
Moves to zero: _____



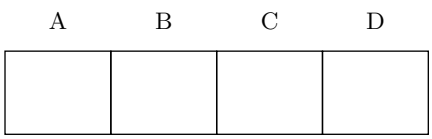
Moves to zero: _____

Problem 3: Choose heights from 0 to 4. Find a start that lasts as many moves as you can; count the move that makes all four towers 0.



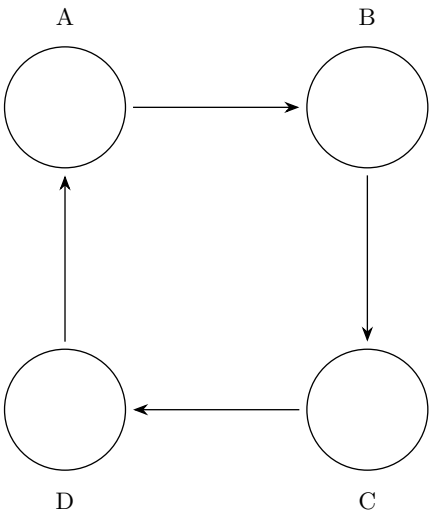


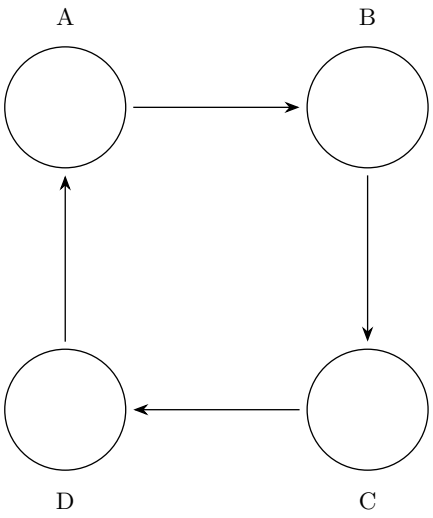
Moves to zero: _____



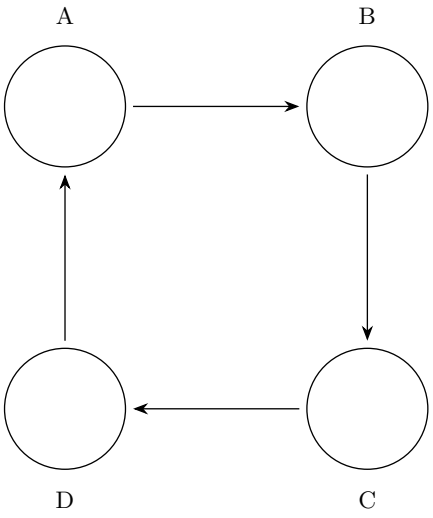
Moves to zero: _____

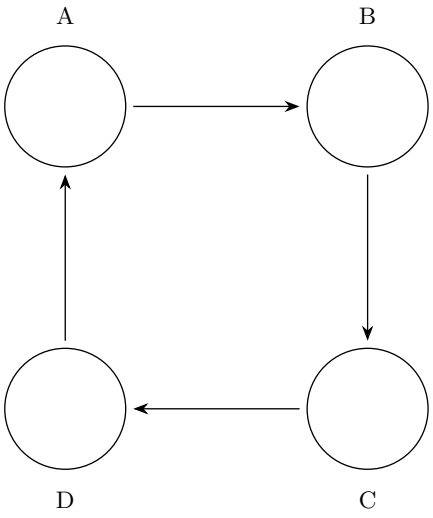
Problem 4: Find different starts that become all zeros in one move.



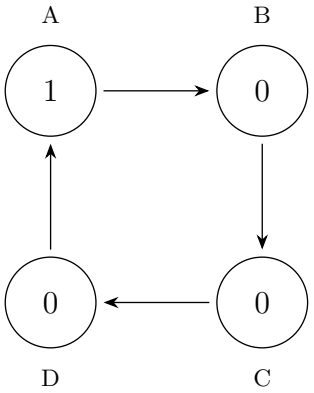


Problem 5: Find starts that take exactly two moves to become all zeros.

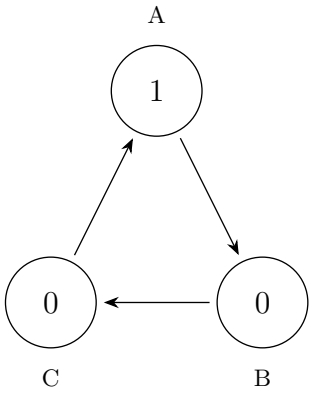




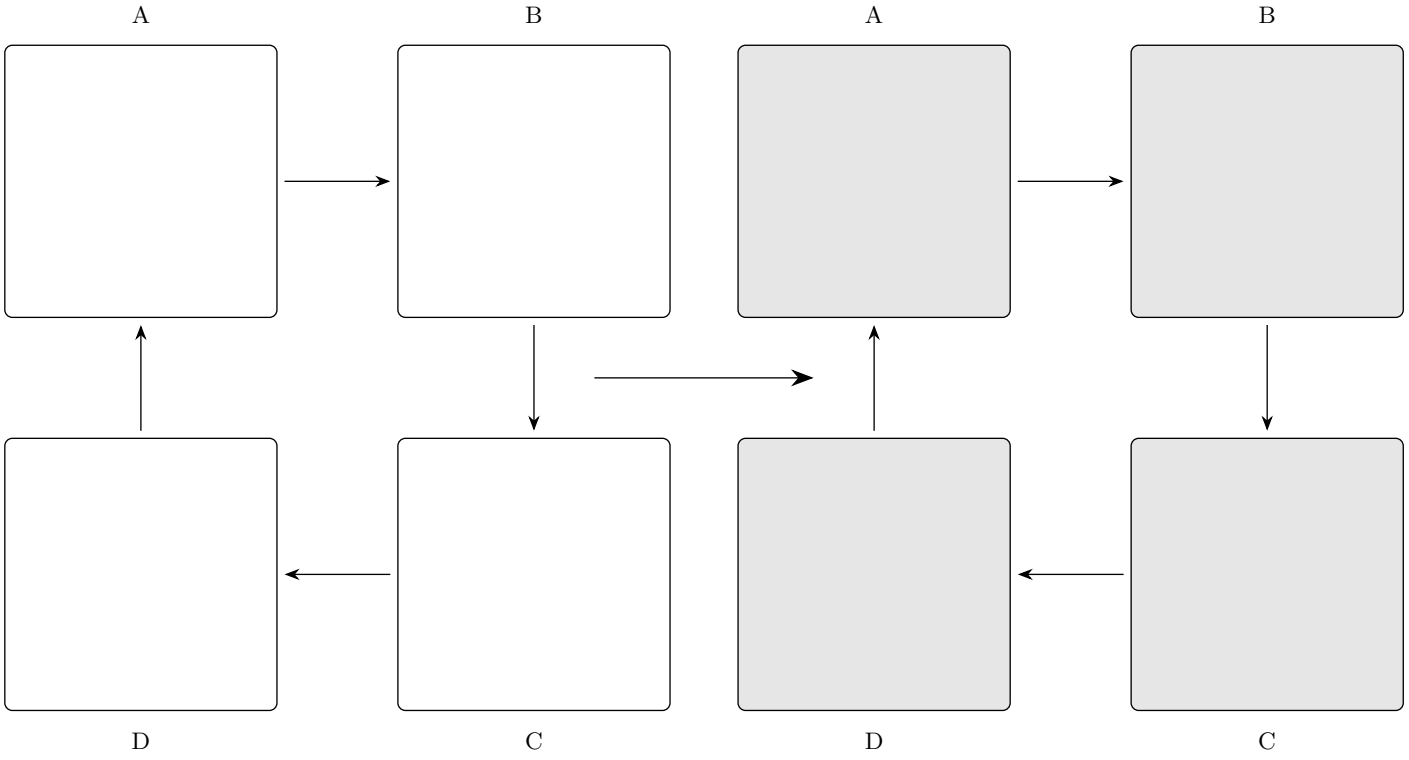
Problem 6: Try both starts with the same gap rule. Stop at all zeros or when the whole ring repeats.



old

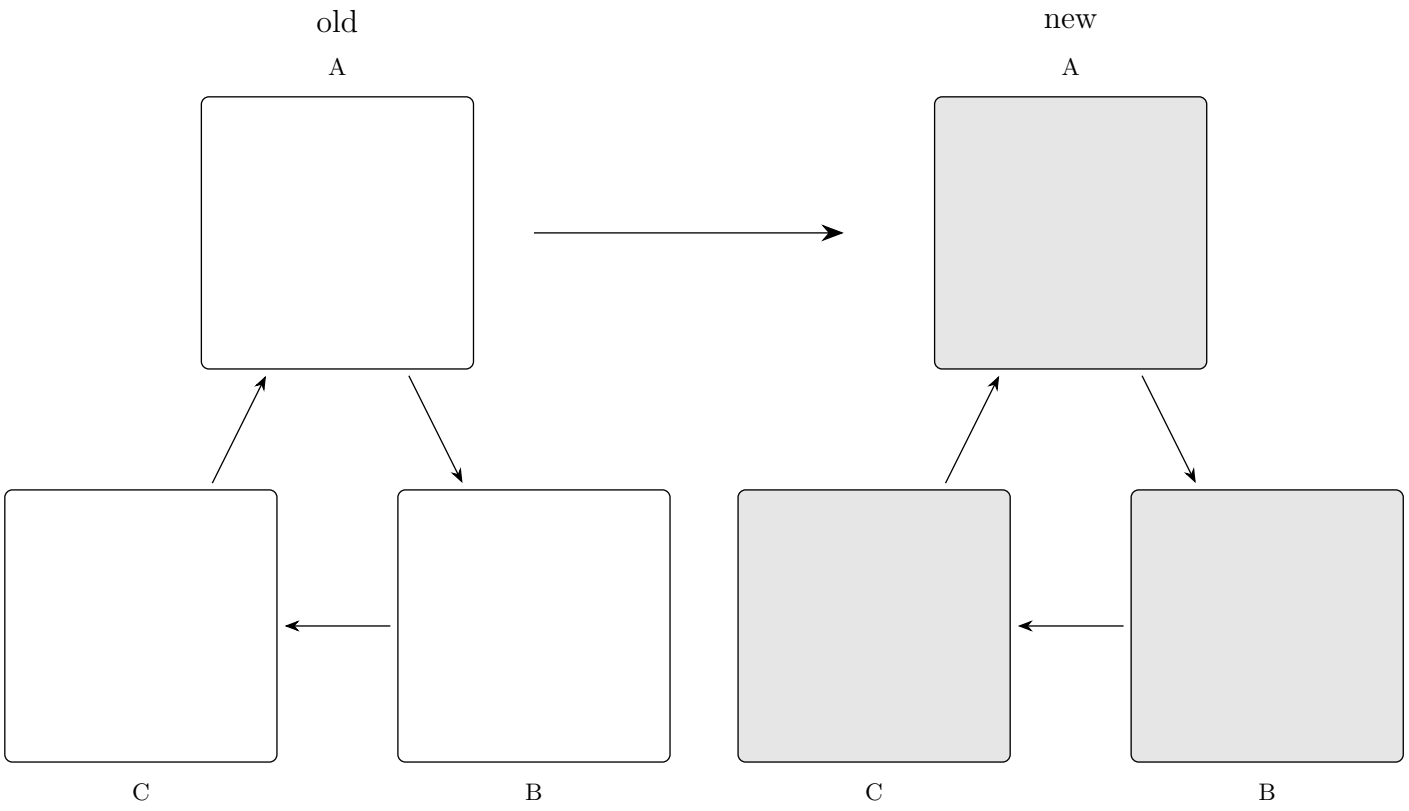


new



	A	B	C	D
1				
2				
3				

	A	B	C	D
4				
5				
6				



	A	B	C
1			
2			
3			

	A	B	C
4			
5			
6			